

HAN ZHENG

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EDUCATION

Johns Hopkins University

Bachelor of Science

Baltimore, MD

Expected May 2026

- Mechanical Engineering (Aerospace Track)
- GPA: 3.79

SKILLS

Design & Manufacturing: SolidWorks (PDM), PTC Creo, Onshape, GD&T, CNC Machining, 3D Printing, Composites Manufacturing

Simulation & Analysis: Abaqus, ANSYS (Mechanical, ACP, CFX, Fluent, Granta), Creo Simulate, SimScale, Micro-CT

Electronics, Programming & Tools: MATLAB, C/C++, Arduino IDE, Circuit Analysis & Implementation, Microsoft Office Suite

TECHNICAL EXPERIENCE

Chief Engineer

Johns Hopkins University - Design/Build/Fly (DBF)

Baltimore, MD

September 2023 - Present

- Placed 17th in the world at the 2026 DBF competition (previously 49th), achieving the best finish in team history and the first full completion of all missions since 2011
- Led the design and fabrication of a 12 lb carbon fiber aircraft with high payload capacity for the 2026 DBF competition
- Simulated a 2-ft hard-landing scenario on a composite fuselage structure using ANSYS ACP and Explicit Dynamics, identified high-stressed regions in the fuselage skin, and applied carbon fiber plies locally for reinforcement
- Completed an aerodynamic trade study in ANSYS CFX evaluating landing gear wheel fairings, confirming a ~5% drag reduction and a net performance benefit over weight-induced drag penalties
- Manufactured the aircraft fuselage using a CNC-milled XPS foam mold with a 3-layer carbon fiber layup ($\pm 90^\circ$, $\pm 45^\circ$, $\pm 90^\circ$), and fabricated primary airframe components including wings, tail components, and control surfaces using wet layup
- Managed a 12-member engineering team, coordinating design reviews, task allocation, composite manufacturing, and integration for aircraft development
- Performed structural analysis on the aircraft wing using a self-developed MATLAB-based beam analysis tool to optimize fuel tank placement, reducing maximum bending moment by 16%

Mechanical Engineering R&T Intern

Albany Engineered Composites

Rochester, NH

June 2025 - August 2025

- Investigated the relationship between fiber crimp and elastic properties of 3D-woven composites, identifying a positive linear correlation ($R^2 = 0.66$) between crimp and in-plane moduli
- Devised and conducted experiments to characterize friction in woven preforms across varying temperature, moisture, and architectural conditions, establishing its role in compaction response and fiber buckling behavior
- Developed a Fortran subroutine linking strain and modulus from test data, enabling Abaqus simulation of a micro-CT-compatible preform compaction fixture to assess stress and deflection under compaction loads
- Designed a fixture in SolidWorks to secure woven preforms during micro-CT scanning and 3D-printed it in ABS, replacing a foam fixture prone to sample movement and compromised scan quality

PROJECT EXPERIENCE

Structure Lead

Mechanical Engineering Senior Design Project

Baltimore, MD

September 2025 - Present

- Designed and integrated a structural interface adaptor between the MRI test bed and a hydraulically actuated motion platform
- Engineered a three-degree-of-freedom (3-DOF) motion system consisting of dual linear stages and a rotational stage, each guided on 1/2" Garolite rails and integrated with hydraulic piston actuation for controlled motion
- Performed extensive structural simulations in ANSYS to evaluate stage bending, iteratively optimizing geometry to achieve over 5 $\times$ reduction in deflection

Fabrication Engineer

Johns Hopkins University Whiting School of Engineering

Baltimore, MD

September 2024 - December 2024

- Created detailed process sheets for manufacturing Stirling engine components (flywheel, piston housing, base plate) from CAD drawings, incorporating GD&T to define tolerances for each manufacturing step
- Operated lathe, mill, and other tools to fabricate a Stirling engine running at approximately 200 rpm